

PORSCHE AG (P911.DE) vs. FERRARI N.V. (RACE.MI)

Relative Value Diligence: A Structural Value Trap, Not a Statistical Arbitrage

Field	Detail
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Classification	Relative Value / Pairs Analysis — Automotive & Luxury Goods
Coverage Universe	Porsche AG (XETRA: P911) · Ferrari N.V. (Borsa Italiana: RACE / NYSE: RACE)
Sample Period	September 30, 2022 – September 21, 2026 (1,008 trading days, live market data)
Recommendation	NO ACTION — reversion thesis not statistically supported; monitor, do not initiate
Time Horizon	Not applicable — thesis rejected pending re-test at next quarterly catalyst
Prepared For	Portfolio / technical work sample — asset management application
Model Repository	[GITHUB REPOSITORY LINK — insert repository URL, e.g. github.com/gayathritvinod/porsche-ferrari-quant-framework]

Report Structure

- 1. Executive Summary & Investment Thesis
- 2. Fundamental Business Model Divergence
- 3. Econometric Framework — Log Ratio, ADF, Chow, Ornstein-Uhlenbeck
- 4. 3-State Probability-Weighted Valuation Engine & Sensitivity Grid
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- 7. Methodological Appendix — Notation, Data Sources & Disclosures

Disclosure. *This document is a self-directed technical research exercise prepared for portfolio and interview purposes. It is not investment research produced by or on behalf of a regulated broker-dealer, does not constitute investment advice or a solicitation, and should not be relied upon for any investment decision. All company data is sourced from public disclosures and market data providers as cited in Section 7.3; scenario financial assumptions in Section 4 are explicitly illustrative and are marked as such throughout.*

1. Executive Summary & Investment Thesis

This report re-underwrites the Porsche AG / Ferrari N.V. relative-value pair using a full econometric and fundamental diligence process, live market data through September 21, 2026, and a five-module statistical framework purpose-built to stress-test a reversion thesis rather than assume one. **The conclusion is unambiguous: the P911/RACE price ratio exhibits the quantitative and fundamental signature of a structural value trap, not a mean-reverting statistical arbitrage, and we recommend no position be initiated on a reversion thesis at this time.**

1.1 Two Competing Hypotheses

Naive mean reversion treats the historical average of the P911/RACE price ratio — in this case, a pooled since-IPO average near 0.25 — as a fixed gravitational center the ratio must eventually return to, independent of whether the underlying businesses generating that ratio have themselves changed. **Structural value trap** describes the alternative: the relationship between the two businesses has genuinely and durably shifted, such that the old average no longer describes anything real, and a position sized on reversion to it will not be rescued by the passage of time.

1.2 Summary of Findings

Test / Module	Live Result	Verdict
Stationarity (Augmented Dickey-Fuller)	Statistic -1.2043 ; $p = 0.6717$	Fails — cannot reject random walk
Structural Break (Chow Test)	$F = 1,081.01$; $p < 0.0001$	Confirmed — pre-2024 regime invalidated
Reversion Speed (Ornstein-Uhlenbeck)	$\kappa = 0.4013/\text{yr}$; half-life 435d; 95% CI $(-0.23, 0.98)$	Point estimate plausible; not statistically resolved
Current Dislocation (Regime Z-Score)	$Z = -0.38$ (post-decoupling regime)	No actionable stretch today
Historical Backtest (6m / 12m)	13.8% / 13.8% hit rate; -4.59% / $+1.60\%$ avg. return	Fails — historically unprofitable
Fundamental Cross-Check (Scenario Engine)	Model ratio 0.076–0.092 vs. market 0.125 vs. original target 0.25	Corroborates the statistical read

1.3 Bottom Line

Every independent line of evidence in this report — econometric, statistical, and fundamental — points the same direction. The ADF test cannot establish mean reversion in the post-2024 data; the Chow test confirms, with overwhelming statistical force, that the regime underlying the original 0.25 reversion target no longer exists; the Ornstein-Uhlenbeck reversion-speed estimate carries a confidence interval that spans zero; the one fully model-free test in the framework — the historical backtest — shows this exact setup has failed roughly six times out of every seven it has occurred; and an independent, bottom-up fundamental valuation converges on an equilibrium ratio below where the pair trades today, not above it. We are not recommending a mean-reversion trade on this pair, and we view the original thesis's 0.25 reversion target as a statistical artifact of a regime that broke in mid-2024.

The remainder of this report develops each of these findings in full: Section 2 establishes the fundamental business divergence underlying the statistical break; Section 3 presents the econometric framework in complete methodological detail; Section 4 develops an independent, bottom-up valuation cross-check;

Section 5 subjects the reversion thesis to a model-free historical backtest; Section 6 sets out the forward monitoring framework; and Section 7 documents notation, data sources, and limitations.

2. Fundamental Business Model Divergence

The statistical break identified in Section 3 does not occur in a vacuum. Both businesses have undergone genuine, disclosed operational and strategic divergence since the original thesis was framed in February 2026 — divergence that independently explains why the pre-2024 relationship no longer holds, and why an eventual reversion to it should not be assumed.

2.1 The Market Has Already Re-Priced Both Names

Metric	Feb 2026 (Original Thesis)	Sept 2026 (Current)	Change
P911 share price	€41.61	€45.12 ¹	+8.4%
RACE share price (Milan, EUR)	€311.70	€360.50 ²	+15.6%
Implied P911/RACE ratio	0.133	≈ 0.125	Marginally lower

This is the central real-time finding of the fundamental review: seven months into what was framed as a 12–18 month reversion call, the ratio has not moved toward the original 0.25 pivot — it has drifted marginally lower even as both stocks rallied in absolute terms, because Ferrari outperformed Porsche on the way up as well as, historically, on the way down. That asymmetric outperformance is the signature of a **structural regime shift**, not a mean-reverting spread coiling before release, and it is corroborated independently by the stationarity test in Section 3.2.

2.2 Margin Trajectory: Restructuring Drag vs. Guidance Beat-and-Raise

	Porsche AG	Ferrari N.V.
H1 2026 actual	Group operating margin recovered to 7.8%, up from the 0.2% trough over Q1–Q3 2025 cited in the original thesis ⁴	Q2 2026 EBITDA margin of 39.0% ⁵
FY2026 guidance	VW Group cut consolidated operating margin guidance to “up to 1%” (from 4.0–5.5%) after a €6bn non-cash Porsche goodwill impairment, part of roughly €10bn in one-off charges; ex-charges margin ≈ 4% ³	Guidance raised to ≥39% EBITDA margin, ≥€7.60bn revenue, adjusted diluted EPS ≥€9.68 ⁵
Trajectory	Sequential recovery off a trough, against a structurally lower full-year guidance band and a multi-year restructuring drag (Section 2.4)	Two consecutive quarters of beat-and-raise against an already premium margin base

2.3 Luxury Pricing Power vs. Order-Book Moat

Ferrari's order book “now entirely covers 2027” per CEO Benedetto Vigna's Q2 2026 remarks ⁵ — longer than the “late 2027” coverage the original thesis cited seven months earlier, indicating the order-book moat has widened, not merely held steady. The original thesis's average-selling-price CAGR calculation conflated three distinct effects that this report separates explicitly: (i) like-for-like pricing on an unchanged model, the strict definition of pricing power; (ii) product-mix shift (for example, the roughly €400k+ Purosangue displacing the roughly €200–250k Roma/Portofino in the volume mix); and (iii) personalization revenue, which represents approximately 20% of car revenue and is a monetization lever distinct from list-price increases. Only the first of these three is pricing power in the strict sense used in a relative-value framework; the latter two are

volume-and-mix effects that happen to land in the same average-selling-price line and materially overstate underlying pricing power if left unseparated.

2.4 EV Transition Risk — A Scored Framework

Porsche has confirmed approximately 9,000 cumulative job cuts through 2035 (5,000 direct reductions at Zuffenhausen/Weissach, 1,900 announced previously, and 2,000 non-renewed contracts), underwritten by a €2.1bn restructuring commitment and employee pay/bonus concessions exchanged for 2035 job guarantees⁴ — a multi-year, structurally embedded cost base, not the one-off charge the original thesis's €2.7bn estimate implied. Ferrari's Luce EV drew a mixed-to-controversial design reception but sold out its deliberately constrained approximately 500-unit 2026 allocation, and the company is reported to be doubling down on internal combustion⁶ — hedging EV exposure through engineered scarcity rather than betting the model line on broad battery-electric reception, which materially de-risks the single catalyst the original thesis treated as its most important swing factor.

Risk Factor (1 = low, 5 = high)	Porsche	Ferrari	Rationale
% of 3-yr volume/revenue exposed to BEV	4	1	Macan/Taycan lines are BEV-dependent; Luce allocation is ~500 units against ~14,000 total deliveries
BEV gross margin vs. ICE gross margin	5	2	Porsche's BEV economics are dilutive amid the China price war; Luce is priced at a premium (~€500k+), preserving margin
Restructuring cost as % of market cap	4	1	~€2.1bn+ committed vs. ~€41bn market cap (~ 5%); Ferrari has disclosed no comparable charge
Composite score (average)	4.3 — High	1.3 — Low	

3. Econometric Framework

This section presents, in full methodological detail, the four-part statistical test battery used to evaluate whether the P911/RACE price ratio behaves as a mean-reverting series. Each subsection states the governing equation, defines every symbol, and reports the exact live-data result.

3.1 The Log Price Ratio

All downstream testing operates on the natural log of the price ratio rather than on raw prices or a simple ratio, because log differences are additive and symmetric over time — the property every regression-based test in this section requires to be valid.

$$r_t = \ln (P911_t / RACE_t)$$

3.2 Stationarity: The Augmented Dickey-Fuller (ADF) Test

The ADF test asks whether the log ratio is stationary (mean-reverting) or contains a unit root (a random walk with no pull-back force). The test regresses the day-to-day change in the series on its own prior level:

$$\Delta X_t = \alpha + \beta X_{t-1} + \varepsilon_t$$

Symbol	Definition
ΔX_t	The day-to-day change in the log price ratio (today's value minus yesterday's value).
α	A constant drift term.
β	The coefficient of interest: a significant negative β indicates a real pull-back force toward the mean (mean reversion); β statistically indistinguishable from zero indicates a random walk.

Symbol	Definition
X_{t-1}	The prior day's level of the log ratio — how far from average the series already stood.
ϵ_t	Unpredictable daily noise.

Null hypothesis: $\beta = 0$ (the series is a random walk with no reversion force). The test statistic and its associated p-value are computed against Dickey-Fuller's non-standard critical value tables, not an ordinary t-distribution.

Live result: ADF statistic = -1.2043 ; p-value = 0.6717 . **At $p = 0.6717$, we cannot reject the null hypothesis of a unit root at any conventional confidence level. Statistically, the P911/RACE log ratio has not been shown to be mean-reverting.**

3.3 Regime Stability: The Chow Structural Break Test

The Chow test asks whether the relationship governing the log ratio changed at an identifiable point in time — specifically, mid-2024, coincident with Porsche's disclosed China-demand deterioration and EV-strategy reset (Section 2). It compares the leftover error from one trend line fit across the full sample against two separate trend lines fit before and after the candidate break date:

$$F = [(SSR_{pooled} - SSR_{split}) / k] / [SSR_{split} / (n - 2k)]$$

Symbol	Definition
SSR_{pooled}	Leftover squared error from fitting one trend line across the entire sample.
SSR_{split}	Combined leftover squared error from fitting two separate trend lines, split at the candidate break date.
k	Parameters per line (2: intercept and slope).
n	Total observations across both periods.
F	Ratio of error improvement per added parameter to remaining error per remaining degree of freedom — large F indicates the two-line fit is not explainable by chance.

Live result:
F-statistic = $1,081.01$;
p-value < 0.0001 (break date: June 1, 2024). **This is one of the most statistically decisive results in the framework. The pre-2024 regime that produced the original thesis's 0.25 pooled average is**

Symbol	Definition
confirmed, with overwhelming statistical force, to be a different regime from the one the pair trades in today.	

3.4 Reversion Dynamics: The Ornstein-Uhlenbeck (OU) Process

Having established (Section 3.2) that stationarity is not statistically confirmed, the OU process is nonetheless fit to quantify what reversion speed and equilibrium level would be implied if a pull-back force exists, and — critically — how precisely that speed can actually be estimated from the available history:

$$dX_t = \kappa(\theta - X_t) dt + \sigma dW_t$$

Symbol	Definition
X_t	The log price ratio at time t .
θ (theta)	The long-run equilibrium level the series is pulled toward.
κ (kappa)	Annualized speed of reversion — how hard the pull back to θ is per unit of distance.
σ (sigma)	Diffusion / volatility of the random shocks along the way.
dW_t	A standard Brownian increment (pure random noise).

Half-life — the time for a deviation from θ to close by half — follows directly from κ :

$$t_{1/2} = \ln 2 / \kappa$$

Kappa and theta are not observed directly; they are recovered from an AR(1) regression of today's log ratio on yesterday's, with the regression's slope and intercept converted to κ and θ via closed-form algebra. Because a single historical path is only one realization of many the data-generating process could have produced, the regression is additionally re-estimated on 1,000 bootstrap resamples (drawn with replacement) to build an honest confidence interval around κ and the implied half-life, rather than relying on one point estimate.

Parameter	Live Estimate
κ (speed of reversion, annualized)	0.4013 / yr — 95% bootstrap CI: (-0.2318, 0.9769)
θ (equilibrium, log-ratio units)	-2.3875 → ratio level 0.0919
Half-life	435 trading days (≈ 20.7 months) — 95% bootstrap CI: (176, 4,691) trading days
Bootstrap draws implying $\kappa \leq 0$ (no reversion)	12.1% of 1,000 resamples

Interpretation. The κ point estimate implies an economically plausible ~21-month half-life, but its 95% confidence interval spans zero and extends into negative territory —

Parameter	Live Estimate
meaning “no reversion at all” remains statistically consistent with the same data. With approximately 1,008 trading days of history against a ~435-day half-life, the sample spans only ~2.3 half-lives — mathematically insufficient to pin down κ precisely, regardless of how plausible the central estimate appears. This estimate does not override the ADF and Chow findings above; it should be read alongside them, not in place of them.	

4. 3-State Probability-Weighted Valuation Engine & Sensitivity Grid

Independent of the time-series framework in Section 3, this section develops a bottom-up fundamental cross-check: an EV/EBITDA bridge to a per-share price for each company under bear/base/bull assumptions, probability-weighted to a single expected value, and stress-tested across a two-way sensitivity grid.

4.1 Methodology

Each scenario applies an EBITDA growth and margin adjustment, together with a scenario-specific exit EV/EBITDA multiple, to a common baseline (Porsche: €6,860m EBITDA, 19.4% margin, €4,100m net debt, 911m shares; Ferrari: €2,970m EBITDA, 39.0% margin, €2,000m net debt, 180m shares — illustrative baseline figures; see Section 7.4). Enterprise value is computed as EBITDA × exit multiple; equity value as EV less net debt; and price per share as equity value divided by shares outstanding. Scenario probabilities of 25% / 50% / 25% (bear / base / bull) are applied to compute a single probability-weighted expected value per name.

4.2 Scenario Output

Scenario	Prob.	P911 Price (€)	RACE Price (€)	Implied P/R Ratio
Bear	25%	19.09	331.64	0.058
Base	50%	32.87	432.38	0.076
Bull	25%	51.34	532.86	0.096
Expected Value	—	34.04	432.32	0.076

4.3 Valuation vs. Prevailing Market Price

	Market Price (Sept 2026)	Model Expected Value	Gap
Porsche (P911)	€45.12 ¹	€34.04	Market at a ~32.6% premium to model fair value
Ferrari (RACE)	€360.50 ²	€432.32	Market at a ~16.6% discount to model fair value

This is the sharpest independent cross-check in the report: the market is already priced more consistently with Ferrari re-rating upward and Porsche re-rating downward than with the P/R ratio reverting up toward the original 0.25 target. The model's expected ratio of 0.076 — and the OU-implied equilibrium of 0.092 from Section 3.4 — both sit **below** the current market ratio of ≈ 0.125 , not above it. Two independent methodologies, built on entirely different data and assumptions, converge on the same conclusion: the ratio's fundamental “home” has moved to roughly a third of the original 0.25 target, and it may have further to compress rather than room to recover.

4.4 Two-Way Sensitivity Grid — Porsche Implied Price (€)

EBITDA Growth ↓ / Exit EV/EBITDA →	3.5x	4.5x	5.0x	5.5x	6.5x	7.0x
–15%	17.9	24.3	27.5	30.7	37.1	40.3
–5%	20.5	27.7	31.3	34.8	42.0	45.6
0% (base)	21.9	29.4	33.2	36.9	44.4	48.2
+5%	23.2	31.1	35.0	39.0	46.9	50.8
+15%	25.8	34.5	38.8	43.1	51.8	56.1

4.5 Interpretation: Operating Leverage vs. Order-Book Moat

The grid's cross-term reveals a compounding, not merely additive, risk structure for Porsche: EBITDA-growth sensitivity is approximately €37 of implied price per unit of growth at a 5.0x exit multiple, but rises to approximately €52 per unit of growth at a 7.0x multiple — an operating miss and a multiple de-rate are not independent risks here, they compound each other. This is by design in the scenario construction: Porsche's exit multiple range (3.5x–7.0x, a 50% bear-to-bull ratio) is far wider in percentage terms than Ferrari's (22x–32x, a 69% bear-to-bull ratio, per the original thesis's scenario assumptions). That asymmetry is the quantitative expression of a qualitative reality documented in Section 2: Ferrari's multi-year order book anchors its valuation multiple even in a stress scenario, while Porsche — a capital-intensive, cyclical, China-exposed OEM mid-restructuring — faces the prospect of an earnings miss and a valuation de-rate arriving together, not in isolation.

Limitation. The EBITDA, margin, net debt, and exit-multiple assumptions underlying this section are illustrative, calibrated to the original February 2026 thesis, and are not current sell-side consensus figures. Treat Sections 4.2–4.5 as a directional cross-check corroborating the statistical findings of Section 3, not as a precise price target; refresh with current consensus fundamentals before relying on these figures for a live investment decision.

5. Historical Backtesting & Risk Analytics

Sections 3 and 4 are both, in different ways, parametric: they assume a model form (an AR(1)/OU process; an EV/EBITDA bridge) and ask whether the data is consistent with it. This section applies a fully model-free test — a direct historical backtest — that asks only whether the trade this thesis implies has actually worked in the past, with no distributional assumption of any kind.

5.1 Tracking Error & Information Ratio

Tracking error measures the annualized volatility of the daily return difference between the two names — the risk actually borne while waiting for convergence:

$$TE = \sigma(r_t^{(P911)} - r_t^{(RACE)}) \times \sqrt{252}$$

The Information Ratio then expresses expected payoff per unit of that risk:

$$IR = E[\alpha] / TE$$

Metric	Live Value
Annualized tracking error	33.22%
Information Ratio (on the original thesis's illustrative 38.4% target alpha)	1.16

Interpretation. An IR of 1.16 would, on the framework's own scale, support a “high conviction” label ($IR > 0.5$). This figure should be read with caution: it is computed against the original thesis's illustrative 38.4% target alpha, which implicitly assumes convergence upward toward the legacy 0.25 target. Sections 3.4 and 4.3 both independently indicate the fundamentally and statistically supported equilibrium sits below today's ratio, not above it — meaning the alpha assumption feeding this IR's numerator has itself been undermined by the same analysis that produced it. This IR should not be quoted as a standalone conviction signal until the alpha assumption is recomputed against the live scenario output in Section 4.

5.2 Out-of-Sample Historical Backtest

For every historical date the rolling Z-score entered a “moderately cheap” band (-2.0 to -1.5), we measure forward performance at 6- and 12-month horizons. A “hit” requires the ratio to move in the expected direction and close at least 50% of the original gap to its long-run mean.

Horizon	Episodes	Hit Rate	Average Return
6-month	29	13.8%	-4.59%
12-month	29	13.8%	+1.60%

5.3 Interpretation: The Model-Free Verdict

A 13.8% hit rate requires no model to interpret: a coin flip would produce a 50% hit rate, and the framework's own threshold for a “genuinely recurring” pattern is a hit rate above 60%. Historically, entering this exact dislocation has failed roughly six times out of every seven, with a negative average return over the following six months. This is the single most damning result in the report precisely because it carries no model risk of its own — it is a direct empirical record, independent of whatever the Ornstein-Uhlenbeck point estimate in Section 3.4 suggests.

6. Catalyst Calendar & Monitoring Triggers

A rejected reversion thesis is not a closed file — it is a position to keep monitoring against explicit, dated triggers. This section separates confirmed corporate calendar events from the statistical re-test rules that would be required to revisit the conclusions of this report.

6.1 Confirmed Corporate Calendar

Date	Event	Relevance	Priority
Oct 7, 2026	Porsche AG Capital Markets Day ⁷	Updated margin-recovery trajectory and 2027–29 restructuring milestones	High
Oct 12, 2026	Porsche AG Pre-Close Call, Q3 ⁷	Early directional signal ahead of the formal quarterly statement	Medium
Oct 27, 2026	Porsche AG Q3 2026 Quarterly Statement ⁷	Underlying margin actuals vs. guidance; China revenue trend; EV capex cadence	High
Nov 3, 2026	Ferrari N.V. Q3 2026 Results ⁸	Order-book coverage update; EBITDA margin vs. raised guidance; Luce delivery data	High

6.2 Statistical Monitoring Triggers

Trigger	Rule
Half-life breach rule	If the realized ratio has not closed at least 50% of the gap to θ (ratio level 0.0919) within 435 trading days of this report's data date (September 21, 2026 — implying a re-test checkpoint around mid-2028), treat the reversion thesis as falsified pending full review, not as “still working.”
Two-signal anomaly gate	Do not treat a new persistent $ Z \geq 2$ breach (20+ trading days, Section 3) as investable on the price ratio alone; require confirmation from a second, independent signal (for example an EV/EBITDA-based Z-score) before escalating conviction.
Quarterly K re-estimation	Re-run the OU bootstrap each quarter. Escalate conviction in a reversion thesis only if the 95% confidence interval for K moves decisively above zero across at least two consecutive quarters.
EV transition re-score	Re-run the 3-factor EV transition risk composite (Section 2.4) at each Porsche capital markets day or quarterly release. Escalate if Porsche's composite score falls below 3.0 (from 4.3 currently).

6.3 Qualitative Catalysts to Monitor

- China luxury-auto demand trajectory and its read-through to Porsche's volume mix and BEV pricing pressure.
- Cadence and reception of further Luce deliveries beyond the initial ~500-unit allocation, as a read on Ferrari's broader EV optionality.
- VW Group capital allocation signals affecting Porsche AG, given VW's controlling stake and the FY2026 guidance cut.
- Financing-cost environment for capital-intensive, restructuring-heavy Porsche relative to cash-generative, low-capex-intensity Ferrari.

7. Methodological Appendix

7.1 Consolidated Notation

Symbol	Definition	Used In
r_t	Log price ratio, $\ln(P911_t / RACE_t)$	§3.1
$\Delta X_t, X_{t-1}$	Day-to-day change in the log ratio; the prior day's level	§3.2 (ADF)
$\alpha, \beta, \varepsilon_t$	ADF regression intercept, slope, and residual noise	§3.2 (ADF)
$SSR_{pooled}, SSR_{split}$	Leftover squared error, one pooled trend line vs. two split lines	§3.3 (Chow)
k, n, F	Parameters per line; total observations; the Chow F-statistic	§3.3 (Chow)
κ (kappa)	Annualized OU speed of reversion	§3.4 (OU)
θ (theta)	OU long-run equilibrium level	§3.4 (OU)
σ (sigma), dW_t	OU diffusion volatility; a standard Brownian increment	§3.4 (OU)
$t_{1/2}$	Half-life: $\ln(2) / \kappa$	§3.4 (OU)
$TE, IR, E[\alpha]$	Annualized tracking error; Information Ratio; expected excess return	§5.1

7.2 Model Equations, Consolidated

Log price ratio:

$$r_t = \ln (P911_t / RACE_t)$$

Augmented Dickey-Fuller regression:

$$\Delta X_t = \alpha + \beta X_{t-1} + \varepsilon_t$$

Chow structural break F-statistic:

$$F = [(SSR_{pooled} - SSR_{split}) / k] / [SSR_{split} / (n - 2k)]$$

Ornstein-Uhlenbeck SDE and half-life:

$$dX_t = \kappa(\theta - X_t) dt + \sigma dW_t \quad t_{1/2} = \ln 2 / \kappa$$

Tracking error and Information Ratio:

$$TE = \sigma(r_t^{(P911)} - r_t^{(RACE)}) \times \sqrt{252} \quad IR = E[\alpha] / TE$$

7.3 Data & Sources

Price data: Daily closing prices for P911.DE and RACE.MI, September 30, 2022 through September 21, 2026 (1,008 observations), sourced via the Yahoo Finance data feed (yfinance), adjusted for splits and dividends.

Fundamental & corporate data:

- ¹ Porsche AG (P911) key metrics — stockanalysis.com
- ² Ferrari (RACE.MI) share price — [Yahoo Finance](https://finance.yahoo.com)
- ³ VW Group slashes 2026 profit forecast on €6bn Porsche writedown — [CBT News](https://www.cbts.com)
- ⁴ Porsche to cut another 5,000 jobs — [electrive.com](https://www.electrive.com)
- ⁵ Ferrari Q2 2026 results, guidance raised — [StockTitan](https://stocktitan.com)

- ⁶ Ferrari Luce orders extend into late 2027, doubles down on ICE — [Yahoo Autos](#)
- ⁷ Porsche AG financial calendar 2026 — [Porsche AG Investor Relations](#)
- ⁸ Ferrari N.V. 2026 corporate calendar — [StockTitan](#)

Software & libraries: Python 3, NumPy, pandas, SciPy, statsmodels (Augmented Dickey-Fuller implementation), yfinance. Full model source is available at the repository link on the title page.

7.4 Limitations & Disclosures

- The Section 4 valuation engine's EBITDA, margin, net debt, and exit-multiple inputs are illustrative assumptions calibrated to the original February 2026 thesis, not refreshed sell-side consensus figures; Sections 4.2–4.5 should be read as a directional cross-check, not a price target, until refreshed.
- The Ornstein-Uhlenbeck half-life estimate (Section 3.4) is subject to material small-sample estimation risk: the available history spans only ~2.3 estimated half-lives, which is the direct cause of the wide bootstrap confidence interval reported.
- The Section 5.1 Information Ratio uses the original thesis's illustrative target alpha and has not yet been recomputed against this report's own Section 4 scenario output; treat it as provisional pending that reconciliation.
- This document is a self-directed technical research exercise prepared for portfolio and interview purposes. It is not a research product of a regulated broker-dealer, is not investment advice, and should not be relied upon for any investment decision.
- This report reflects a live-data snapshot as of September 21, 2026. Because data vendors continuously revise historical adjusted closing prices, a later re-run of the accompanying script will reproduce the same statistical and fundamental conclusions but may show minor decimal drift in figures such as the ADF p-value, Chow F-statistic, or total observation count — an expected consequence of vendor-side data revision, not a change in methodology.